International Society of the Learning Sciences

Towards an Empirical Understanding of Epistemic Data Agency for Middle School Students During a Data Storytelling Summer Camp

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Abstract: Epistemic data agency highlights learners' capacity to take ownership of shaping their learning process with data, yet little work has examined how it can be operationalized in educational contexts. In this paper, we advance the construct of epistemic data agency by providing a list of indicative actions that demonstrate students' epistemic stances with data. Using video recordings from a two-week middle school summer camp on data storytelling, we conducted case studies of two focal students with contrasting prior data experiences and demographics. Our analysis builds on existing frameworks of knowledge-related epistemic agency activities, which we used to identify activity-level episodes, and extends them by inductively deriving action-level indicators that specify how students engaged with data in practice. This empirical grounding offers researchers and educators an analytical framework for examining how students enact data-related agency and offers a foundation for guiding and evaluating pedagogical interventions to strengthen data literacy education.

Introduction

The concept of epistemic agency was introduced to the field of educational research within the knowledge-building perspective approach to learning (Scardamalia, 2002). It refers to students' abilities to take responsibility for their knowledge-building process by making choices about what to ask, explore, and create. Recent work has spent efforts in developing a more thorough understanding of epistemic agency (Bandura, 1999; Schwartz & Okita, 2004; Palonen & Hakkarainen, 2013; Akgun & Sharma, 2023). Among these efforts, Damşa et al. specifically discriminated between two different categories of indicators of epistemic agency: knowledge-related activities, which specifically focus on idea production as the outcome, and process-related activities that focus on organizing and coordinating the process of knowledge production (Damşa et al., 2010).

While previous studies on epistemic agency are often conducted in broad domains (Stroupe, 2014; Zhan & Louie, 2024; Baze et al., 2025; Chew & LeTourneau, 2025), few have examined how it manifests in contexts where data is the central medium for learning. As data becomes increasingly important to engage with the world (Wolff et al., 2016), understanding how learners perform and develop epistemic agency through data is both timely and necessary. To address the need, existing work by Kahn and Jiang (2024) has explicitly derived the terminology of epistemic data agency as "the conceptual and material practices that reveal and shape learners' relationships to the data." Yet prior work has not offered empirical guidance on how epistemic data agency can be observed or operationalized in practice. In this study, we extend the construct by grounding it in empirical practice. Specifically, we ask: **Which indicators of epistemic data agency can be identified when students work with data in the context of a data storytelling project?**

We conceptualize epistemic data agency as a domain-specific instantiation of knowledge-related epistemic agency, enacted when the medium of knowledge work is data. We derived the framework of epistemic data agency in the setup of a two-week Data Visualization and Environment (DVE) summer camp, where students learn data collection and data visualization and develop stories of personal interest about their communities supported by data. We applied Damşa et al.'s (2010) framework of knowledge-related epistemic agency to video recordings of two focal students' working sessions on their data storytelling project, and then inductively extended it into an empirical framework of epistemic data agency. The two focal cases, each representing students with distinct backgrounds, illustrate how learners can develop and manifest epistemic data agency in diverse ways throughout the project. The contribution of this study is **an empirical framework that identifies and organizes student actions that are indicative of epistemic data agency**. This framework offers researchers and educators an analytical tool to recognize, analyze, and support how students exercise epistemic agency in relation to data.

Conceptual framework: Epistemic data agency

Epistemic agency broadly refers to learners' capacity to take ownership over knowledge-building processes by making choices that shape what counts as legitimate knowledge, how disciplinary practices are enacted, and whose perspectives are valued (Miller et al., 2018; Keifert et al., 2018). It situates learners as knowledge producers

(Damşa et al., 2014) rather than participants in the productions or inquiries of others. Within data-intensive learning environments, this notion extends into epistemic data agency, which Kahn and Jiang (2024) conceptualize as the practices through which learners reveal and shape their relationships to data. Enacting epistemic data agency involves not only performing "data moves" such as selecting variables, modeling, or encoding (Kahn & Jiang, 2021), but also interpreting, critiquing, and reimagining data stories (Radinsky, 2020).

Prior studies have highlighted the importance of fostering such data-related agency and designed learning environments that enable students to act as decision-makers in their data work. For example, Walker et al. (2025) show how youth in data science workshops exercised ownership over data collection and analysis through open-ended projects, while Hardy et al. (2020) reframed students as data producers whose emergent goals, material resistances, and tool use shaped their sense of agency in making and interpreting data. Related efforts, such as storytelling with community datasets (Wilkerson & Polman, 2020), embodied exploration of census data (Roberts & Lyons, 2020), and collaborative inquiry community-relevant projects (Deahl, 2014), similarly highlight how students' roles as active knowledge producers can guide data work. However, despite these advances, few studies have explicitly measured epistemic data agency, leaving open the question of how to systematically identify and analyze it in practice. Addressing the gap, we operationalize this construct at the level of observable actions that learners take in the flow of data work. In contrast to the primarily conceptual account, our contribution is to provide an empirical framework that specifies how epistemic data agency can be identified in practice.

Much prior work frames epistemic agency as a collaborative achievement of groups engaged in collective knowledge building (e.g., Scardamalia, 2002; Damşa et al., 2010). While we acknowledge the social and distributed nature of data work, our analysis foregrounds the individual–data relation as the primary unit of analysis. Rather than examining collaboration itself, we focus on how individual students work with data and how their concrete actions reveal and shape their epistemic stances toward data within the collaborative project context. We are specifically interested in how students' knowledge-related activities with data, such as searching, sharing, structuring, and producing data-based ideas, reveal their epistemic stances. In contrast, process-related activities (e.g., planning, monitoring, coordinating) are treated as important collaborative practices but are beyond the scope of this study, as they pertain more to group regulation than to individuals' agentic relations with data. Therefore, in this study, we only focus on the knowledge-related epistemic agency (Damşa et al., 2010) and investigate how individual students conduct actions that shape their relationship to data. By foregrounding epistemic data agency at the individual knowledge-related level, our analysis examines how students' engagements with data reflect and shape their positions as active knowledge producers in data-rich learning environments.

Method

This IRB-approved study was conducted during a two-week fully-funded summer camp, with the goal to empower students to use real-life data to communicate ideas and prepare them for future engagement with authentic data. During the camp, students learned data visualization design, used digital tools such as Tableau to create charts, developed a project that addresses a self-selected community-related question through data visualizations, and presented their project to parents and community stakeholders. A case study approach (Yin, 2009) was chosen as the data analysis method, given the need for an in-depth understanding of the construct analyzed.

Participants

Sixteen middle school students enrolled in the summer camp, with fifteen providing assent and parental consent to participate (Age_{Ave} = 12.3 years; 6 identified as female and 1 as non-binary). Although 70% reported some prior experience using data, most had not participated in formal data-related programs. For this case study, we first selected two project groups with well-developed final projects. Both groups examined crime-related data but pursued different questions and produced distinct outcomes, enabling a comparison of how epistemic data agency developed. From these groups, we selected one focal student from each to reflect variation in demographic and prior experience with data. Morgan (pseudonym), an incoming 7th-grade female who self-identified as "Black, African, Caribbean, or African American" and "White, Caucasian, or European American," reported no prior experience with data. Omar, an incoming 9th-grade male who self-identified as "Arab, Middle Eastern, or North African," reported experience in four data-related practices, including using spreadsheets, creating graphs, working with data in science projects, and using chart creation tools, placing him on the higher end of the cohort.

Educational context

Summer camp setup

The study was conducted during a DVE summer camp in 2025 on an R1 university campus in the US. The summer camp was designed as a collaborative, project-based learning experience and was offered to middle school

students from a local district. Activities were led by a science teacher from the students' school with assistance from the research team, including helping with troubleshooting and questions answering. During the first week, students formed into groups of 2-3 students and identified their project topics as they practiced data visualization skills through structured problems. For the second week, students primarily worked with their groups to develop their projects, with facilitators offering group and individualized scaffolding as needed. On the last day of the camp, students presented their work to parents and community stakeholders in posters or tri-boards at a showcase.

Data storytelling project

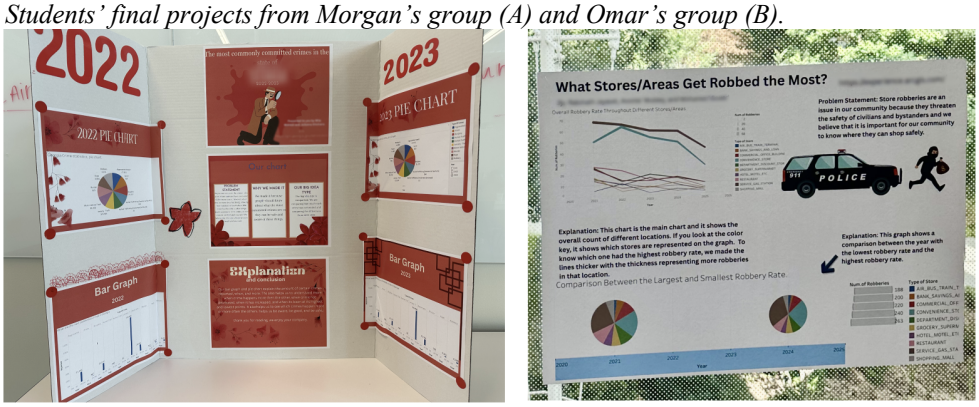
The final project of the summer camp was divided into four design phases: 1) identify the topic and data, 2) plan out the project, 3) create the visualization, and 4) compose the story, where students decide topics and make decisions collaboratively while the camp facilitators provide guidance.

Morgan, together with another incoming 7th-grader, chose the topic of the most commonly committed crimes in their home state. They constructed their dataset based on the state government's annual crime data report, including the number of different crimes reported each year. After creating the visualizations using Tableau, they used a tri-board and put individual charts and support materials on the board (see Figure 1.A).

Omar, together with two other incoming 9th-graders, conducted a project on which stores are robbed the most in the students' home city. They constructed their dataset from the open data portal of the city's police department, which provides detailed reports of crimes occurring in the city, including the location and time of each incident. This group organized their materials on a poster (see Figure 1.B).

Figure 1

Students' final projects from Morgan's group (A) and Omar's group (B).



(A) (B)

Data and analysis

All camp sessions, including the instructions and students' working sessions, were video recorded and transcribed. In addition, students answered a pre-survey about their prior experience with data and data visualization, and a post-survey self-reporting their experience and growth of skills throughout the summer camp. Finally, students participated in focus groups with 5-6 students each time, which were audio-recorded and transcribed. Students' worksheets created during the camp and the final projects were also collected.

In this work, we focus on the video data. Video analysis provides rich and fine-grained access to students' interactions with data, enabling us to capture both verbal reasoning and embodied moves (Atkins et al., 2009; Derry et al., 2010). We examined the working sessions of two selected students across project phases (Morgan: 403 minutes; Omar: 407 minutes of video recordings). Our analysis looks for both students' activities and actions. We use Jordan & Henderson (1995)'s definition of activities as goal-oriented stretches of interactions that are typically marked by a recognizable purpose or set of tasks, such as students trying to understand what the two axes mean in a graph. Actions, in contrast, are observable and discrete components of activity that the students conduct that help them move towards the goal of their activities (Damşa et al., 2010), such as searching "y-axis" in Google to find the definition, or drawing a sketch of the x-axis and the y-axis and confirming with teammates.

We conducted a two-stage qualitative analysis. In the first stage, we deductively identified activities by applying Damşa et al.'s (2010) framework of knowledge-related activities: searching information, sharing ideas, structuring ideas, and producing ideas. For each project phase, we identified video episodes in which the focal student engaged in these four categories of activity, accompanied by detailed descriptions of what the student said and did in sequence. The first author initially segmented a subset of the video data into episodes and discussed the segmentation and descriptions with the research team weekly. After reaching an agreement, the same criteria

were applied to the full dataset and reviewed with the team, resulting in 82 episodes for Morgan and 92 for Omar , with the length from 30 seconds to 6.5 minutes). In the second stage, for each category of activity in each project phase, we inductively coded the descriptions of the identified activities into finer-grained actions that represent the discrete and observable moves the student made (e.g., entering a new data point and clarifying a terminology), with codes first derived by the first author and then iteratively refined through team discussion before being applied to the full dataset. These identified actions serve as empirical indicators of epistemic data agency, specifying how students enacted the broader knowledge-related activities in the context of working with data. In this way, our contribution builds on an existing framework of activities while extending it with empirically grounded indicators of action.

Results

Table 1 summarizes our findings of indicative actions of students' epistemic data agency observed in each of the project phases across the four activity categories based on Damşa et al. (2010). While students performed actions spanning all four activity categories during each phase, we have highlighted selected activities that demonstrate significant differences in how the focal students manifested epistemic data agency. Morgan and Omar demonstrated distinct yet evolving forms of epistemic data agency. Morgan's engagement was often exploratory, and sense-making oriented, while Omar's was more assertive and decision-driven, reflecting differences in prior experience with data. Across phases, students' agency also shifted from exploring and negotiating ideas (Phases 1–2), to coordinating collaborative work (Phase 3), and finally to synthesizing data narratives for communication (Phase 4). The following sections illustrate these patterns through selected episodes from each project phase.

Table 1

Indicative actions of students' epistemic data agency across the four activity categories.

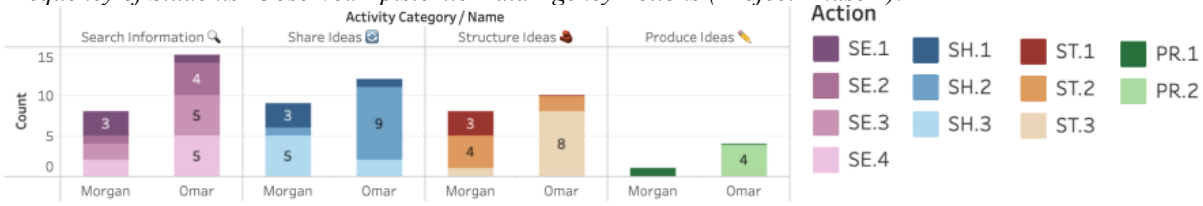
Activity Category (Damşa et al., 2010)	Indicative Action (Design phase where the action was identified)
Searching information (SE)	**SE.1** Search for external datasets to extend or verify existing data (Phase 1) **SE.2** Refine search strategies for data accuracy (Phase 1) **SE.3** Investigate raw dataset (Phase 1, 3, 4) **SE.4** Evaluate dataset relevance (Phase 1) **SE.5** Refer back to prior resources for guidance (Phase 2, 3) **SE.6** Search for visual references to inform data representation designs (Phase 2, 4) **SE.7** Search for external definitions to clarify meanings or data context (Phase 3)
Sharing ideas (SH)	**SH.1** Express and negotiate understanding of data concepts (Phase 1, 2) **SH.2** Propose and justify data or design decisions (Phase 1, 2, 3) **SH.3** Identify and Communicate data insights (Phase 1, 2) **SH.4** Build on or critique teammates' ideas (Phase 2, 3) **SH.5** Use analogies or personal references to explain ideas (Phase 3) **SH.6** Coordinate team decisions to ensure data coherence (Phase 3, 4) **SH.7** Frame the communicative purpose of data (Phase 4)
Structuring ideas (ST)	**ST.1** Formulate guiding questions for data use (Phase 1, 2) **ST.2** Experiment with data organization strategies (Phase 1, 3, 4) **ST.3** Define and validate dataset boundaries (Phase 1) **ST.4** Connect data to real-world contexts (Phase 2) **ST.5** Generate and compare alternative data solutions (Phase 2, 3) **ST.6** Refine the dataset for accuracy (Phase 3) **ST.7** Coordinate design and data consistency (Phase 4) **ST.8** Integrate prior knowledge into current data work (Phase 4)
Producing ideas (PR)	**PR.1** Compare and commit to a dataset (Phase 1) **PR.2** Reorganize the dataset into a new structure (Phase 1) **PR.3** Integrate data and design ideas across artifacts (Phase 2, 3) **PR.4** Summarize conceptual decisions into expressions (Phase 2) **PR.5** Compare individual work to generate a combined idea (Phase 3) **PR.6** Synthesizing data narrative (Phase 4) **PR.7** Refining design based on project goals (Phase 4)

Phase 1: Identify the topic and data

During this project phase, students work with teammates to identify the project topic and the dataset(s). They are provided with a reference list of datasets, but may also search online for other resources. The goal is to construct a dataset on a Google sheet for chart creation in Tableau. **Figure 2** presents how Morgan and Omar enacted epistemic data agency across activity categories, showing how students with different prior experiences engage with data practices to make sense of, justify, and organize information in their personally meaningful ways.

Figure 2

Frequency of Students' Observed Epistemic Data Agency Actions (Project Phase 1).



When conducting information searching, Morgan demonstrated the *Look for external data resources* action (**SE.1**) when she expressed a need to *"*get some inspiration*"* before she expanded her search for more specific neighborhood crime datasets. She conducted a broader searching strategy by trying out different searching keywords and looking into multiple data resource links. In contrast, Omar engaged in the same action by extending his exploration through tracing embedded external links from provided resources, such as opening an interactive crime map and locating new data. While Morgan broadened her search to new information channels, Omar deepened the utility of existing resources by exploring their linked contents.

When sharing ideas, Morgan frequently enacted the *Express and negotiate understanding of data concepts* action (**SH.1)**. After looking through the dataset, she frequently asked herself, *"*what's a dataset?*"* and *"*how do I get the data?*"* Omar more often engaged in the *Propose and justify data or design decisions* action (**SH.2)**. For example, Omar justified the dataset choice to the facilitator and teammates by demonstrating how it could be filtered by location to focus on local data, and even suggested using his teammate's neighborhood to make the analysis more relatable. As he examined the dataset, he decided to focus on the crime categories with more cases ("we're definitely gonna do the top ones") and advocated deleting those with fewer cases. This contrast suggests that while Omar, as a student with more prior experience with data, enacted epistemic data agency through assertive pitching and justification, Morgan, as a student with few prior experiences with data, engaged in a more cautious, confirmatory form of sense-making when working with data.

While both students engaged in the *Experiment with strategies to organize data* action (**ST.2**) when structuring ideas, they enacted it in distinctly different ways. Omar approached his data manipulation with a goal-oriented mindset, actively manipulating filters and searching for neighborhood lists to refine the dataset around a specific comparative logic: "[Neighborhood 1] is rich and [Neighborhood 2] is (not) so there can be a contrast." His actions showed intentional control over how data should be organized to reveal meaningful data patterns, demonstrating epistemic authorship in structuring the dataset to support interpretation. Morgan, in contrast, took a more exploratory approach. Rather than reorganizing existing data, she first explored how datasets could be created in different ways, and expressed excitement towards different data ("this is scary," "I'm actually interested in this"), and experienced both directly downloading data or creating her own dataset ("*I get a free dataset but I can make one.*") Omar approached data organization as a way to pursue a defined analytical goal, whereas Morgan used it as an opportunity to explore what data could be and how it might be constructed, highlighting how students can enact epistemic data agency in different yet complementary ways.

Phase 2: Plan out the project

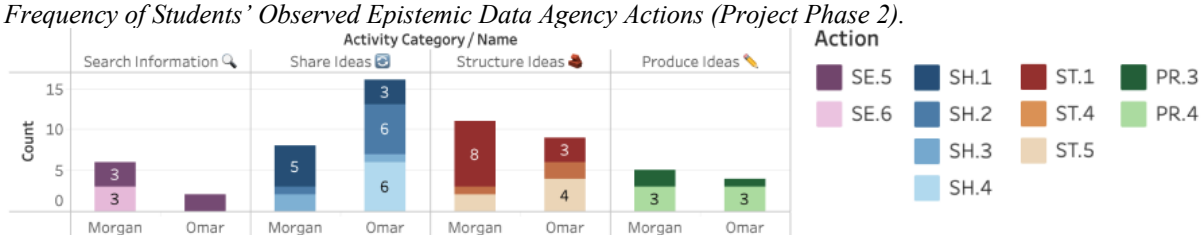
During Phase 2, students follow the DPV (Domain, Purpose, Visual) worksheet designed in prior work (Li et al., 2025) to guide their data visualization design. The framework helps middle school students plan their data stories by defining (1) Domain: the problem statement, the potential audiences of their story, and the conventions or unspoken rules in this area, (2) Purpose: the communication goal of their data story, and (3) Visual: how data are mapped to visual elements. The output is a filled DPV worksheet that they could follow in their later design. **Figure 3** presents how the students enacted epistemic data agency when planning the project. While Morgan's agency mainly manifested through questioning and conceptual synthesis, Omar's agency emerged through critical evaluation and leadership in design execution.

When sharing ideas, in this phase, students mainly shared and debated their ideas to guide their team's design choices. Morgan mainly enacted *Express and negotiate understanding of data concepts* (**SH.1**), frequently

questioning and clarifying basic visualization ideas such as axes, color, and position ("am I right that this bar stands on the x-axis?"). She often explained her reasoning process to her teammate, even sketching a bar chart to illustrate her understanding. On the other hand, Omar enacted *Build on or critique teammates' ideas* (**SH.4**), taking an active role in shaping team decisions. He compared design options (e.g., color and chart type), refined ideas through discussion ("the size of the line varies depending on the number of crimes"), and persuaded teammates to focus on creating a poster instead of a tri-board. While Morgan focused on making sense of visualization concepts, Omar focused on leading the team to refine and act on their design ideas.

Figure 3

Frequency of Students' Observed Epistemic Data Agency Actions (Project Phase 2).



For the structuring idea activity, students collaborated by discussing, comparing, and refining how to represent data in their project. Morgan mainly enacted *Formulate guiding questions for data use* (**ST.1**) by asking clarifying and reflective questions such as "why is this important to our community?" and "is convention like unspoken rules?" Her questions refined the project's conceptual scope and guided her team's understanding of the data. In contrast, Omar more often engaged in *Generate and compare alternative data solutions* (**ST.5**), taking a more decision-oriented approach. He compared chart types ("line chart or pie chart") and discussed how visual variables like size could represent data differences ("the size of the line varies depending on the number of crimes"). He also initiated debates about project format ("I hate tri-boards… posters make more sense") and justified his preferences with prior examples from the summer camp. These differences suggest that epistemic data agency in structuring ideas can manifest both as conceptual inquiry and as design negotiation.

When producing ideas, students moved their thinking and discussion forward by turning emerging interpretations into concrete project ideas and recording their decisions. Morgan showed a stronger enactment of epistemic data agency (she conducted five actions during this Phase 2 and only one action in Phase 1), indicating her growing ownership in connecting data and design. She engaged in *Integrate data and design ideas across artifacts* (**PR.3**) by linking earlier worksheets to the current DPV planner and deciding how to visualize data through color and chart type. She also enacted *Summarize conceptual decisions into expressions* (**PR.4**) by turning team discussions into concrete choices, such as coloring bars by category and proposing conventions like "percentage" or "different types." Omar contributed to his group by refining the team's worksheet and ensuring prior statements were accurately transferred. Together, their work demonstrates how students began transforming their data reasoning into coherent visual expressions.

Phase 3: Create the visualization

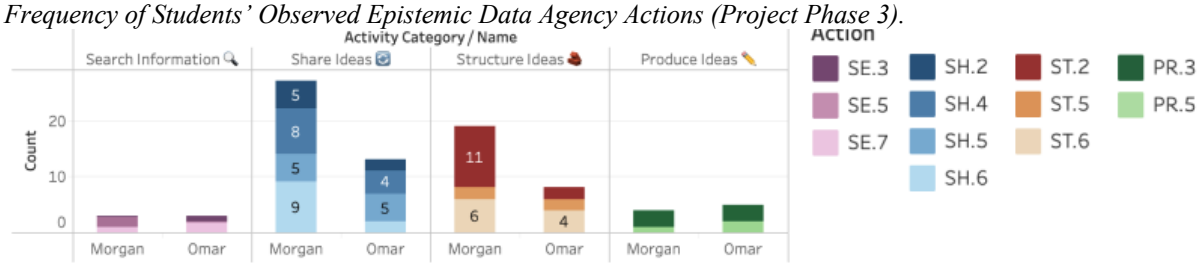
During the visualization creation phase, students first sketched up their planned data visualizations, drawing on the ideas outlined in their project planner. They then worked in Tableau to construct charts. The outcome of this phase is a set of charts created in Tableau that reflect the design students developed during Phase 2. **Figure 4** presents how Morgan and Omar enacted epistemic data agency when creating their charts. Overall, students' epistemic data agency evolved from exploring and proposing ideas to more coordinating and sustaining collaborative work. Specifically, Morgan's agency became increasingly organizational and leadership-oriented, while Omar's shifted toward maintaining group coherence through supportive reasoning and validation.

During this phase, Morgan exhibited a clear growth in her epistemic data agency, especially through her active participation in *Build on or critique teammates' ideas* (**SH.4**) and *Coordinate team decisions to ensure coherence between data and design* (**SH.5**). She moved from asking conceptual questions in earlier phases to taking an organizing and guiding role in team decision-making. Morgan frequently monitored the group's progress, clarified task sequences ("we sketch first, then move to Tableau"), and ensured alignment between their data use and visual representation. She built on teammates' ideas by questioning their choices ("did you add (all the values) together in the pie chart?") and reasoning through alternatives, such as whether to include both raw counts and percentages to serve different viewer perspectives ("we should do both, what if someone wants to see it in percentage"). When technical or design issues arose, she reviewed her teammate's work, explained the rationale for dataset decisions, and suggested adjustments to improve readability. However, upon examining the pie chart that she and her teammate created, the research team found that it did not accurately represent the dataset, as the

proportions of the slices did not correspond to the actual data values. This finding suggests that, while Morgan actively engaged in sharing and structuring ideas, she did not recognize the inaccuracy in their visualization, indicating an ongoing need to strengthen her data literacy skills. Nevertheless, her proactive involvement in coordinating the team's reasoning and design decisions reflects a growing epistemic data agency, demonstrating emerging leadership and deeper engagement with data practices, even though such agency does not necessarily equate to advanced technical proficiency in data visualization.

Figure 4

Frequency of Students' Observed Epistemic Data Agency Actions (Project Phase 3).



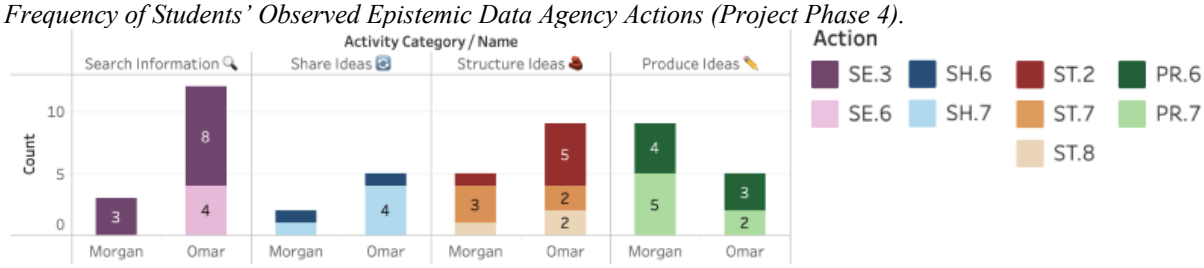
On the other hand, while Omar continued to demonstrate epistemic data agency through collaborative meaning-making and refinement of visual representations, particularly in Sharing Ideas activities and Structuring Ideas activities, the number of such actions decreased compared to earlier phases (for example, Omar conducted **SH.2** six times in Phase 2 but only twice in Phase 3). He engaged actively in peer discussions such as clarifying definitions ("hostage is robbery"), constructing imagined scenarios to reason through ambiguous terms, and encouraging teammates to verify or adjust their sketches and charts ("I will have year1 year2 year3 on the x-axis"). His participation centered more on negotiation and validation of teammates' ideas than on initiating new directions, as seen when he offered supportive suggestions such as "just size it down bro" to resolve chart issues. These actions reflected sustained engagement in reasoning and coordination but were less generative and self-directed than before, signaling a shift from leading analytical exploration to collaboratively maintaining group coherence, which is a more supportive enactment of epistemic data agency.

Phase 4: Compose the story

During this phase, students organized their charts created using Tableau into a poster or a tri-board, and added decorations and supportive text to their data story using the software Canva. The outcome of this phase is the data story that would be presented in the showcase. **Figure 5** presents how students enacted epistemic data agency across different activity categories at Phase 4. In this phase, students' epistemic data agency culminated in more integrated and purposeful forms of engagement as they synthesized data reasoning with the communication purpose. Morgan demonstrated increasing self-direction and ownership, integrating diverse ideas into her team's final product. Omar, after a temporary decline in agency enactment during Phase 3, re-engaged through data-grounded actions that reconnected analytical reasoning with design execution.

Figure 5

Frequency of Students' Observed Epistemic Data Agency Actions (Project Phase 4).



In this final phase, Morgan showed clear growth in the Producing Ideas activity (with nine observed actions in Phase 4 compared with one action during Phase 1) and became increasingly self-directed. In the *Synthesizing Data Narrative* (**PR.6**) action, Morgan showed an emerging capacity to integrate data accuracy, narrative coherence, and communicative clarity. While adding and resizing bar charts, she cross-checked details with her teammate ("I'm doing 2022 right?") and discovered mismatched years, showing awareness of how the story should be connected with the data. She revisited the project planner to align their "big idea" with what the data actually showed, refining titles and explanations to ensure consistency between evidence and message. Meanwhile, in *Refine Design based on Project Goal* (**PR.7**), she adjusted background colors, layout, and chart

sizes not merely for consistency but to ensure the visual message was clear and accessible to an audience. Her design decisions reflected consideration of how viewers would interpret the information.

Omar, on the other hand, showed a renewed increase in his epistemic data agency actions, especially in Searching Information activity after a dip in Phase 3. For example, in the *Investigate raw dataset* (**SE.3**) action, Omar leveraged the original dataset to refine his search process and achieve more targeted results after several failed online attempts. Rather than continuing broad exploration, he strategically returned to the dataset as a reliable source ("we have grocery, gas station… night club") and used them as keywords to guide subsequent online searches for visual or contextual references. Throughout the poster-building process, he coordinated data and visual layout, suggesting chart placement, resizing, and prioritizing content ("remove the border… to keep key information"). These decisions show how Omar consistently grounded his design in data, reflecting a continued yet more practical and goal-oriented enactment of epistemic data agency in this phase of the project.

Discussion

This study advances an empirical framework that identifies actions through which students enact epistemic data agency from searching and organizing data to structuring and producing data-driven ideas. Mapping these actions across the four project phases, we demonstrate that epistemic data agency involves not only interpreting and visualizing data but also negotiating meaning and design decisions. This framework contributes to data literacy education by offering a lens to analyze how individuals work with data within collaborative contexts. Future work would apply the framework on other students to examine the degree to which learners activate epistemic data agency and to identify how to design curriculum and instructional interventions to effectively support it. This approach can help the field of data literacy education move toward a richer understanding of how epistemic data agency can foster more empowered and authentic participation in data literacy learning.

In our case study, the two focal students have various prior experiences and demographic backgrounds. Analyzing how they work with data following the proposed framework revealed not only how one student's agency developed throughout the project, but also how epistemic data agency can manifest differently across learners. For example, Morgan, who had less prior experience with data, initially expressed and negotiated her understanding through questions, clarifications, and coordination with peers. Omar, in contrast, frequently proposed and justified data and design decisions early in the project, assuming a directive stance in team discussions, but his engagement decreased during Phase 3 before resurging in the final phase. These patterns suggest that epistemic data agency manifests dynamically, shaped by learners' prior experiences, self-efficacy, and ongoing roles within collaboration. The framework thus serves as a useful analytic lens to identify not only whether students enact epistemic data agency, but also to what extent and in what ways it emerges, offering a foundation for designing future interventions that support diverse learners. While the current work did not prioritize explaining these factors in-depth to explain the reasons for these differences, future research could explore the influencing factors, such as students' backgrounds, self-efficacy, and collaboration dynamics, and investigate personalized strategies to activate epistemic data agency across diverse learners (Damşa et al., 2010).

While prior work on epistemic agency often emphasizes its collaborative nature (e.g., Scardamalia, 2002; Damşa et al., 2010), this study centered on epistemic data agency as features of individual students: how a learner takes ownership of interpreting, representing, and communicating data in the context of a group context. Focusing on the individual level does not conflict with collaborative perspectives; rather, our findings reveal how individual actions intersect with shared epistemic processes. For example, Morgan's questions elicited teammates' clarifications, while Omar's design justifications prompted group reasoning about communicative intent. Prior research in collaborative learning similarly suggests that epistemic work often emerges through the interplay between individual contributions and shared meaning-making processes (e.g., Stahl, 2006; Suthers, 2006). These interactions demonstrate that individual epistemic acts can serve as entry points for collective sense-making. Future work could examine how individual and collective agency co-evolve in data-rich learning settings and how instructional design can foster both individual and collaborative epistemic engagement.

This study's summer camp context and small number of focal cases might limit the generalizability of the findings. Firstly, the phases in which we identified epistemic data agency from the two focal students are not the only phases where such agency may occur. For other students, the same epistemic action might emerge in different phases or recur across multiple stages of the project. Future research should examine how these indicators manifest across larger and more diverse populations. Additionally, our focus on knowledge-related activities excluded process-related activities, such as goal setting and time management (Scardamalia, 2002). Future studies could integrate process-related indicators to provide a fuller picture of how learners manage and negotiate both individual and shared epistemic goals when working with data. Finally, expanding this framework to different contexts, such as formal education classrooms (Akgun & Sharma, 2023) and informal learning spaces (Letourneau et al., 2025), will help validate its value for designing learning environments to enhance learners' data literacy.

Reference

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Acknowledgements

This material is based upon work supported by the National Science Foundation (NSF AISL #2314109).